SC4024/CZ4124 Assignment #1 - Summary Report

Name: Bhati Nancy **Title**: World Happiness Report **Duration:** 4.59 mins
Matric No.: U2222368G **Video No.:** V12

1. Objectives

The main objectives of this presentation were:

- To visualise global happiness trends from 2015 to 2019.
- To examine how country rankings have changed over time.
- To analyze key factors influencing happiness scores.
- To understand relationship between factors and happiness scores.

2. Novelty

Country rankings based on happiness have always fascinated me. Therefore, through this project I aimed to understand how these rankings are determined, and which factors influence happiness.

I began with a storyline to clearly show global happiness score, top-ranking countries, relationship of key factors with happiness scores and analyze Singapore's happiness.

To achieve this, I used a mix of simple yet effective charts such as line charts, bar charts and scatter plots etc combined with more unique visualisations like stacked bar charts, butterfly charts and geospatial (chloropleth) maps.

- I used a stacked bar chart using Python to compare the top 3 and bottom 3 happiest countries in 2019 across key factors highlight why top countries outperform the rest [1].
- I designed a butterfly chart in Tableau to compare the factor scores of Finland and Singapore. It was a new chart I learned for this project [2].
- I also used a geospatial chart to show the global happiness scores from 2015-2019 [3].
- I learned and created interactive box plots in Python (Plotly) to display happiness scores distribution across years (2015-2019) [4].
- I have also used an interactive radial chart to compare top 6 countries across each happiness factor [5].

3. Technical Challenges and Innovation

I used Tableau and Python for creating visualisation and PowerPoint and Zoom for creating and recording presentation.

Initially, the challenge was to identify the most important tool for specific visualisations. Tableau was chosen for geospatial and comparative charts due to its ease of use while Python was used for box plots, scatter plots and heatmap because of familiarity and precision.

A significant technical challenge was merging multiple annual datasets (2015-2019) into a unified dataset. Working with separate files proved time consuming but after merging, it became more efficient.

Another challenge involved chart selection and tool limitations. My aim was to use simple and understandable visuals. Tableau's box plots did not meet my expectations, prompting the use of Python. This step led to the exploration of interactive visualisation using Plotly.

Furthermore, the project introduced new visual charts for me such as butterfly chart, geospatial charts, interactive box plots and radial chart.

Overall, this project strengthened my technical proficiency in Python, Tableau while also enhancing my ability to apply data visualisation as storytelling with data visualisation principles.

4. References

[1] Dataset used and weblink.
This dataset (world Happiness Report 2015-2019) was taken from Kaggle.
https://www.kaggle.com/datasets/unsdsn/world-happiness

[2] References and weblinks that have contributed preliminary ideas to your visualisation (if available).
https://www.worldhappiness.report/

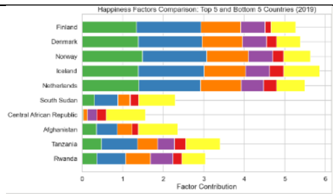
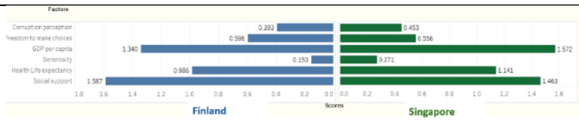
[3] Other references and weblinks (if relevant)…..
Butterly Chart (Tableau) https://youtu.be/-_7yOIjTATU?si=LFQKLvKxNsulg01Y
Box Plot using Plotly in Python https://www.geeksforgeeks.org/python/box-plot-using-plotly-in-python/

[4] My work
- Tableau Visualisation Workbook
- Python (.ipynb)

[6] YouTube Presentation video link: https://youtu.be/z6MR7f5g8e0

[7] Presentation slides

Figures and Tables

1.	Stacked Bar Chart	
2.	Butterfly Chart	
3.	Geospatial Chart	
4.	Interactive Box Plot (Plotly)	
5.	Interactive Radial Chart